F/SOP/FYDP 02/01/00

Department of Mathematics

MT-460 Final Year Design Project

Proposal for the Final Year Design Project

Title	Analyzing Fund Flow Patterns and Market Efficiency in Pakistan with Predictive Models

Domain	Domain 1	Domain 2	Domain 3	Domain 4
	Mathematics	Finance	Data Science	Machine Learning

1. Nature of Project [Tick all that applicable]

☒ New Project OR ☐ Extension of Existing Project	☐ Industrial Collaboration	☐ Funded
☐ Other Department Collaboration (If yes) Department Name_____	☐ Other Academic Institution Collaboration (If yes) Institution Name_____	

2. Brief Outline (*Problem Identification and Significance*)

In the context of the Pakistani financial market, there has been minimal research on index and mutual funds that examines investor sentiment, fund flows, and their impact on market efficiency. Existing literature focuses largely on developed economies, leaving a gap in understanding how these factors interact in Pakistan's unique market environment. This lack of research limits the ability of investors and policymakers to interpret current fund flow trends and their implications for future market movements.

3. Objectives

- Utilize machine learning techniques to predict fund flows.
- Analyze their relationship with market efficiency in the context of Pakistani market.
- By examining historical fund data, the research aims to forecast future investment strategies and determine the optimal weight allocation for each fund in an index.

4. Scope

Pakistani financial market.

5. Proposed Methodology

The methodology for achieving the above objectives begins with collecting relevant index and mutual fund data from the Pakistan Stock Exchange (PSX). The data will then be cleaned and preprocessed to remove inconsistencies and ensure accuracy. Following this, data visualization techniques will be applied to identify patterns, trends, and potential relationships between fund flows, market efficiency, and investor sentiment. Statistical and correlation analysis will be conducted using Excel to quantify these relationships. Subsequently, machine learning models will be implemented to predict future fund flows and derive optimal weight allocations for each fund in an index. Finally, the results will be used to design a data-driven investment strategy tailored to the Pakistani market.

6. Resources Involved

Data sources and Software tools

7. Description of Industrial Support (If any)

8. SDGs (If Applicable)

☐ No Poverty	☐ Zero Hunger
☐ Good Health and Well-Being	☐Quality Education
☐ Gender Equality	☐ Clean water and Sanitation
☐ Affordable and Clean Energy	☒ Decent Work and Economic growth
☒ Industry, Innovations and Infrastructure	☐ Reduced Inequalities
☐ Sustainable Cities and Communities	☐ Responsible Consumption and Production
☐ Climate action	☐ Life Below Water
☐ Life on Land	☐ Peace, Justice and Strong Institutions
☐ Partnerships	

9. Gantt Chart

Year	2025 to 2026										
Months	August 2025	September 2025	October 2025	November 2025	December 2025	January 2026	February 2026	March 2026	April 2026	May 2026	June 2026
Literature Review											
Data Collection											
Data Cleaning & Preprocessing											
Visualization & Trend Analysis											
Machine Learning Model Building											
Results & Investment Strategy											
Report Writing & Final Submission											

10. Details of Project Team

i. Students

No.	Name	Seat No.	Signature (s)
1	Roohan Khan	CF-22029	
2	Laiba Sarfaraz	CF-22019	
3	Aina Batool	CF-22005	
4	Maida Murtaza	CF-22016	

ii. Supervisors / Advisors

	Name	Designation & Department	Address & Contact	Signature(s)
Supervisor	Ms. Ubaida Fatima	Assistant Professor - Department of Mathematics		
Co-Supervisor (If any)				
Industrial Advisor (If any)				

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Project Serial No.: _____ Dated: _____	Signature Convener Steering Committee	Signature FYP Coordinator

☐ Proposal Approved	☐ Not Approved	☐ Returned for Clarification / Modification
Comments: (if any)		

_____

(Signature of Chairperson)

Date: _____